VATSL KRISHNN GOSWAMI

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Interests: Reinforcement Learning, Reward Hacking, AI Control, Learning Dynamics.

EDUCATION

Ashoka University, *B.Sc (Hons.) in Mathematics and Computer Science* Expected May 2027

- Major CGPA: 3.75/4.00 (Magna Cum Laude)
- Relevant Coursework: Reinforcement Learning, Machine Learning, Graduate Decision Theory, Real Analysis, Multivariable Calculus, Abstract Algebra, Linear Algebra, Design and Analysis of Algorithms

Indus International School, *IB Diploma Programme* May 2023

- Scored 42/45 with class rank 2/104. Other Test Scores: SAT: 1560/1600, IELTS: 8.5/9.0

RESEARCH EXPERIENCE

Learning After Reward Hacking: Correction and Transfer | *Independent Research* | Project Report Oct 2026

- Built a Gridworld with a faulty completion checker and trained PPO agents under four histories: shortcut access, narrow legitimate practice, varied navigation practice, and no prior training.
- Measured how training history affected learning on the navigation task after checker repair as well as transfer to a new pickup-and-delivery task, measuring genuine task successes across training checkpoints.
- Used identical demonstrations, policy-head resets, and guided practice to test whether slower learning was better explained by limited navigation practice or effects of prior training on the policy's learned representations.

KL-Regularization & Behavioral Change: What Math Says About the Tradeoff | Technical Blog Sept 2026

- Derived the KL-regularized policy optimum and event-level bounds, quantifying how reference probabilities and reward incentives constrain behavioral change.
- Used the KL chain rule to show why fewer observed failures may not mean the underlying behavior has been corrected; proposed a controlled test of behavioral suppression versus correction.

Research Intern | *Ashoka University* | Guided by Prof. Anirban Sen Mar 2025 – May 2025

- Conducted a cross-sectional analysis of political advertisements on Instagram and Facebook, examining how post sentiments varied near election time using the CrowdTangle API, Python, and SQL.

PROJECTS

MultiAuthLab: A Coding Environment for AI Control | Github

- Designed a reproducible multi-tenant authentication environment as a proposed addition to LinuxArena, with extensible tasks for evaluating AI attack protocols.
- Specified three engineering tasks, cookie migration, account switching, and scoped document export, along with adversarial tasks targeting cross-tenant disclosure and revoked-session writes.
- Built an independent evaluator that replays submitted patches against isolated fixtures, scoring functionality and security separately.

AlphaChess: MCTS & Self-Play Reinforcement Learning *(In Progress)* | Github

- Developed an AlphaZero-style chess pipeline by training a ResNet on Lichess game data. Currently extending it with self-play game generation, replay buffer and an RL training loop.
- Built exact solvers for KQK, KRK, and KPK endgames using dynamic programming and retrograde analysis; compared learned value functions against the Syzygy tablebase solutions.
- Implemented MCTS with PUCT selection and built an evaluation harness for matches against Stockfish across search budgets and training checkpoints.

WORK EXPERIENCE

Founding Engineer | *Beet.Health* | Website Remote | Aug 2025 – Sept 2026

- Built LiveKit voice agents for health-logging workflows. Developed scenario-based evaluations combining tool-call checks with LLM-judged conversations to assess tool-call reliability, escalation, and responses to user prompt injections.
- Handled Product Management, DevOps, and Full-Stack work. Led a team of 5 across engineering and UI/UX; grew the bootstrapped platform to 20 clinics and approximately 600 users.

SELECTED TRAINING

Mathematics Training and Talent Search Programme | *IISER Pune* | Website Pune, MH | May 2025 – June 2025

- One of 30 students selected from 1500+ applicants nation-wide for the fully-funded MTTS programme, which aims to cultivate thinking for Higher Mathematics.
- Completed Level-O coursework in Analysis, Number Theory, and Linear Algebra through interactive lectures emphasizing independent formulation of definitions and proofs.